BURNISH

A mastering suite, module by module: from a mix to a release-ready master



Eighteen modules, one chain, a loudness target

Burnish is a mastering suite in one plugin. You build a **signal chain** out of modules, in whatever order you like: equalisers, compressors and limiters, three adaptive spectral tools, saturation, stereo width, and a **Maximizer** that sets the final loudness. Around the chain sit loudness meters, a gain-matched bypass, dither, presets, undo and snapshots, and **ASSIST**, which listens to the song and builds a chain toward a target for you.

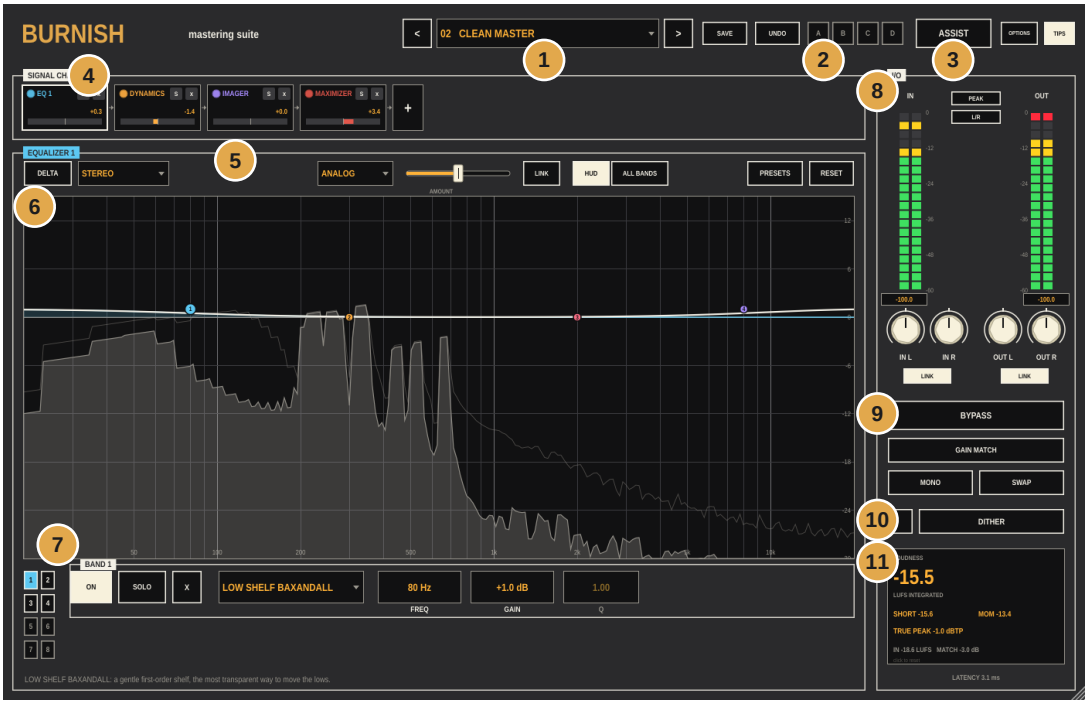
Where Lacquer gives you one knob and decides the rest, Burnish gives you every decision. That is more to learn, so this guide starts with the panel and a first session, then takes the modules one at a time, then dither and delivery. If you are new to mastering, read the first half of the Lacquer guide first: it covers what mastering is for, what LUFS and true peak mean, and why where a song is going changes how loud it should be. This guide assumes that and gets on with the tools.

You'll need

- Ableton Live 11 or 12 (any edition that loads VST3 plug-ins).
- Burnish built and installed. Building copies **Burnish.vst3** into ~/Library/Audio/Plug-Ins/VST3; a release download goes in the same folder.
- A finished mix, and headphones or speakers you know well.

> **The honest bit.** More tools make it easier to do too much. Every module here can make a mix worse as easily as better, and the usual way it goes wrong is that the master gets louder and louder sounds better. Keep **GAIN MATCH** on and press **BYPASS** often. If a change only sounds better louder, it isn't better.

Find your way around



1	Presets	Whole chains, factory and yours: which modules, in what order, with their settings. Click for the list, < > to step. A * means changed since loading. SAVE keeps your own.
2	Undo, snapshots	UNDO (or cmd-Z) steps back. A B C D hold four complete states: shift-click to store, click to recall. Good for comparing two directions.
3	Assist	The assistant: listens, then builds a chain toward a target. See page 5.
4	Signal chain	One tile per module, left to right in signal order. Click to show it, drag to reorder. LED: power. S: solo. x: remove. +: add.
5	Module header	DELTA (hear only what this module changes), the channel mode, the module's own presets, RESET.
6	Module display	The spectrum of the output with the module's curves and handles on it: EQ nodes, crossovers, action regions.
7	Module controls	Everything else for the selected module. A line at the very bottom says in words what it is doing now.
8	Meters, I/O gain	Input and output, peak, RMS or loudness, L/R or M/S. A readout turns red on a clip; click it to clear.
9	Listening	BYPASS, GAIN MATCH (output turned to the input's loudness, so bypass is a fair test), MONO, SWAP.
10	Dither	Power switch and the button that opens the dither panel. Only for 16-bit (or shorter) files.
11	Loudness	Integrated, short-term, momentary and true peak, to ITU-R BS.1770 / EBU R128. Click to restart. The total latency is under it.

Hover over anything for a tip; TIPS turns them on and off for every Burnish. Double-click a knob, fader or readout to reset it. LCD readouts scrub when you drag them up and down. The scroll wheel works on everything.

A first session

1	Prepare	Export your mix as a 24-bit WAV at the session's sample rate, peaking around -6 to -3 dB, with nothing on Live's master. Make a new set, put the WAV on an audio track, and Burnish on the Master.
2	References	Two or three released songs in the same style on other tracks, turned down to match by ear. Solo between them and your track all the way through.
3	Start somewhere	Press ASSIST and let it build a chain (next page), or load a factory preset close to what you want, or start from the empty chain and add modules with +.
4	Listen fairly	GAIN MATCH on. BYPASS every few minutes. Each module's DELTA plays only what it adds: if you can't hear anything in DELTA, the module isn't doing anything worth its latency.
5	Fix, then shape	Problems first (a boomy note, harshness, rumble), then tone and dynamics, then colour and width, loudness last. See ‘Building a chain by hand’.
6	Set the loudness	Maximizer last in the chain. Push GAIN until the loudness block reads what the destination wants, and watch true peak stay under the ceiling.
7	Check elsewhere	MONO. Then a phone, earbuds, a car. Mastering is about all of them, not just your room.
8	Rest and revisit	Come back tomorrow with fresh ears before you send it anywhere.
9	Export	See ‘Dither and delivery’.

Building a chain by hand

Order matters: each module hears what the ones before it did. A common order, and the modules that usually live in each stage:

Stage	Modules	Why here
Clean up	Equalizer 1 (high-pass, narrow cuts), Dynamic EQ, Spectral Shaper, Low End Focus	Problems are easiest to fix before anything else reacts to them. A compressor that is fed a boomy note pumps on it.
Tone and glue	Dynamics or Vintage Compressor, Equalizer 2 or Vintage EQ, Stabilizer, Clarity	Broad tone and gentle compression on a clean signal.
Colour	Exciter, Vintage Tape	Saturation reacts to level and tone, so it comes after they are set.
Space and punch	Imager, Impact, Bass Control	Width once the tone is right; microdynamics and low-end control just before the limiter.
Loudness	Maximizer, or Vintage Limiter	Always last (before dither), so nothing after it can push a peak over the ceiling.

Fewer is better. Three or four modules doing small things beat ten modules each doing a little; and every spectral module adds 43 ms of latency, which Live compensates for but which makes the controls feel slow.

Factory presets

Preset	What it is for
Default	An empty chain. Add modules with +, or run ASSIST.

Preset	What it is for
Clean Master	Gentle EQ, light glue, true-peak limiting to -1 dBTP: a safe pass for most mixes.
Loud and Clear	Clarity and a multiband limiter for a bright, loud master.
Warm Tape	Tape saturation, a program limiter and a touch of vintage EQ for an analogue feel.
Hip-Hop Low End	Bass Control keeps the low end tight and loud; Impact adds punch; Maximizer mode V pushes it.
Electronic Club	Wide tops, mono bass, dense dynamics and a loud, transient-preserving limit.
Rock Glue	Vintage compression for glue, an exciter for bite, a limiter with some character.
Acoustic and Vocal	Light touch: a dynamic EQ to tame harshness, Stabilizer to balance, gentle limiting.
Streaming Ready	Aimed at -14 LUFS with a -1 dBTP ceiling, as streaming services expect.
De-harsh	Spectral Shaper and Low End Focus to clean a mix before anything else.
Vinyl Pre-Master	Mono bass, no clipping, moderate level: something a cutting engineer can work with.
Reference Match	Match EQ against a captured reference, then a clean limit.

Presets set the level of every module, but they can't know your song: the Maximizer's GAIN in particular will need moving. Several modules also have presets of their own, under PRESETS in the module header.

ASSIST

The assistant listens to part of the song and compares what it heard with a target: the **tonal balance** (how the energy is spread across the spectrum), the **loudness**, the **stereo width** in four bands and the **microdynamics** (how much the level moves on a short time scale) in four bands. It then builds a chain to close the gap: Equalizer 1 for tonal balance, Bass Control for the low end, Clarity and Stabilizer for detail and consistency, Impact toward the target's microdynamics, Imager toward its width, and the Maximizer for loudness. Everything it builds is ordinary modules: MODULE VIEW shows them, and they are yours to change.

Setting up a run



The setup page: target, processing, loudness.

Target	Eleven genre profiles, each the typical spectrum, width and dynamics of released music in that style (the LUFS beside each is that style's usual loudness, for reference). + REFERENCE makes a target from an audio file of your own: the assistant finds its loudest eight seconds and learns from those. Click the dot at the left of a target to make it a favourite; the buttons above filter the list.
Processing	Which modules it may use. INTENSITY sets how far it moves things, from SUBTLE (keeps the mix's character) to TRANSFORMATIVE (reshapes tone and dynamics). ANALYSIS is how long it listens, 8 to 60 seconds, or bars of the host tempo with TEMPO SYNC.
Loudness	The loudness to aim for. FULL SCALE sets a -0.1 dB ceiling with true peak off, for a CD or a file nothing will re-encode; STREAMING sets -1 dB with true peak on.
Start	START ASSISTANT, then play the loudest, fullest part of the song, such as the last chorus, until it finishes listening. SAVE AS DEFAULT remembers the setup; ONE-CLICK starts with the saved setup without this page.

The result page



Target library on the left, your song's tonal balance (white) against the target (blue), loudness, and the extras.

EQUALIZER scales every band it set, 0 to 200%. **MAXIMIZER** trims its gain by up to 4 dB either way, and FULL SCALE / STREAMING switch the ceiling. **DYNAMICS MATCH** and **WIDTH MATCH** scale the Impact and Imager bands; **CLARITY** and **STABILIZER** set those amounts. Each has an ON switch for a quick before-and-after. GAIN MATCH and BYPASS are here too. Clicking another target rebuilds the chain from the same listening pass. RELEARN listens again.

> **Two things to know.** After it builds the chain, the Maximizer listens for five more seconds and trims its own gain toward the loudness you asked for, so keep the song playing. And picking a different target rebuilds the modules it uses, which replaces any changes you made to them by hand: choose the target first, then fine-tune. Vocal balance isn't measured, because separating a vocal from the music needs a trained model, which Burnish doesn't include.

The modules

In the order the + menu lists them. Every control also has a hover tip on the panel. Most modules have a **channel mode** beside DELTA: STEREO uses one set of controls for both sides; MID/SIDE and LEFT/RIGHT give each half its own set, A and B, with buttons to switch between them. The multiband modules (Dynamics, Exciter, Imager, Impact) share a **crossover display**: drag a handle to move a crossover or double-click it to type one, hover and click + to add a band, x to remove one; each band has ON and S (solo) at its top; LEARN puts the crossovers at dips in the spectrum of what is playing.

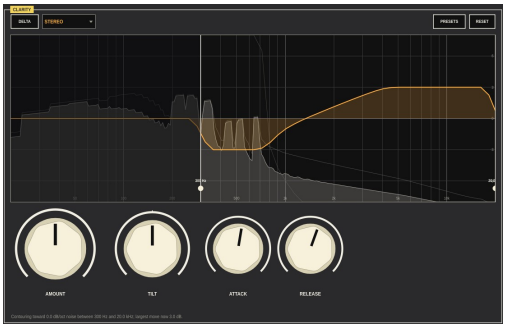
Bass Control



Everything below a CUTOFF (40 to 400 Hz) is the bass it works on; BASS SOLO plays just that. The oscilloscope shows the low band's waveform with the two thresholds across it (ZOOM magnifies it); drag the lines to set them. Two sweet-spot meters show the bass against the rest of the mix and how dynamic it is: aim for green on loud, full passages.

Control	What it does
BALANCE	Level of the low band. The meter above: left lacks bass, right has too much.
PUNCH	Gain on the transients of the low band only: more kick without changing its envelope the way a compressor would. The meter shows the low band's crest factor: left is squashed, right too dynamic.
SUSTAIN POWER	Upward compression below its THRESHOLD (the dashed line): sustained bass is raised toward it, so the low end sounds fuller and steadier. Put the threshold between the kick's peaks and the sustained bass. CHARACTER blends DIRTY (adds harmonics small speakers can reproduce) and CLEAN. AMOUNT sets how much.
PEAK CONTROL	A limiter on the low band at its THRESHOLD (the solid line): put it at the kick's peak so only the loudest hits are held. CHARACTER blends DIRTY (clipping) and CLEAN (look-ahead limiting). Keeps bass peaks from pumping the Maximizer later.

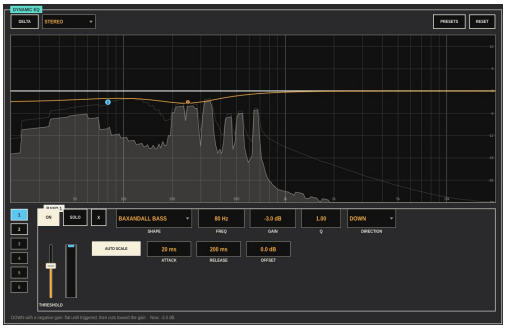
Clarity



An adaptive equaliser with sixty bands that contours the spectrum toward noise with a chosen tilt: resonances come down, holes come up, within 6 dB. It opens up a dull mix without the harshness of a static boost, and can make a master sound louder at the same LUFS. The curve shows the correction now; the handles set where it acts (300 Hz up). 43 ms of latency.

Control	What it does
AMOUNT	How far toward the target. At 100 the most it can move anything is 6 dB.
TILT	The target: 0 dB/oct is pink noise; +3 is white (brighter); -3 is brown (darker).
ATTACK, RELEASE	How fast the contour follows a rise, and how fast it lets go.

Dynamic EQ



Six bands whose gain moves with the level in their own region. A DOWN band with a negative gain is flat until its region gets loud, then cuts: a de-esser, a tamer for one boomy note, a control for harshness that comes and goes. The white curve is static; the amber one moves with the music. Nodes work as in the Equalizer.

Control	What it does
SHAPE, FREQ, GAIN, Q	Baxandall bass or treble shelf, band shelf, bell or proportional-Q bell. GAIN is where the band moves to (or from) when triggered.
THRESHOLD	The level in the band's region that triggers it. The meter beside it shows the region's level outside and the gain change inside.
DIRECTION	UP or DOWN: which way the filter moves when triggered. The line at the bottom of the panel spells out what the current combination does.
ATTACK, RELEASE, AUTO SCALE	How fast it moves and recovers. AUTO SCALE makes low bands slower and high bands faster.
OFFSET	A static gain for the band, whatever the level.

Dynamics

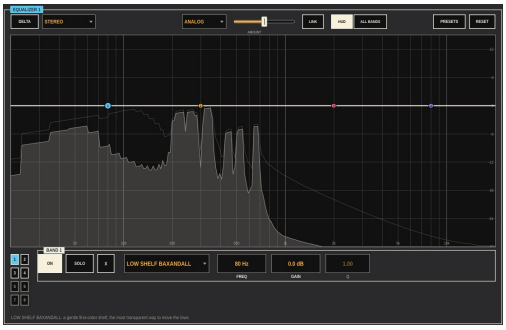


Up to four bands of compression and limiting on the crossover display; each band's line shows its gain change. Each band has a compressor and a limiter on one level detector. The limiter here is the analogue kind, softer than a brickwall: the Maximizer sets the ceiling.

Control	What it does
DETECT	PEAK follows peaks; ENV an averaged level that treats every frequency alike; RMS the average power.

Control	What it does
COMP, LIMIT	The two threshold faders. Above COMP the compressor ratio applies; above LIMIT the limiter's. The meter between them shows the band's input level outside and its gain change inside.
Ratios	Positive ratios compress downward. A negative compressor ratio, shown as (4):1, compresses *upward*: quiet parts rise toward the threshold. A negative limiter ratio expands upward, for punch.
ATTACK, RELEASE, KNEE	Speed in and out, and how gradually it starts. ADAPTIVE REL shortens the release on transients and lengthens it on sustained sound.
PARALLEL, gains	Dry signal mixed back per band; BAND GAIN per band; GLOBAL after all bands. AUTO GAIN matches each band's output level to its input.
Views	CROSSOVER; GAIN TRACE of the selected band; DETECTION, a filter in front of the detector (a HIGHPASS so bass doesn't drive it, or a TILT toward the highs), with S to hear it. ALL BANDS shows every band's key controls; LINK BANDS moves them together.

Equalizer 1 and 2



Eight bands in seventeen shapes, drawn over the spectrum. Drag a node for frequency and gain (shift locks one axis); drag the handles beside a selected node for Q; double-click to add a band or reset a node; alt-click a node to hear only its region; delete removes it; arrow keys nudge (shift coarse, cmd fine). There are two, so one can correct and the other shape.

Control	What it does
Shapes	BELL; PROPORTIONAL Q (narrower the more it boosts or cuts); BAND SHELF (flat top). Shelves: plain, BAXANDALL (gentle first-order), VINTAGE (with a complementary dip), RESONANT (a bump at the corner). Passes: FLAT 6 to 48 dB/oct, RESONANT, BRICKWALL.
ANALOG, DIGITAL	ANALOG: minimum phase, no latency. DIGITAL: linear phase, the same curve with no phase shift, 43 ms of latency.
AMOUNT	Scales every band, 0 to 200%. The grey curve is the unscaled one.
A, B, LINK	In MID/SIDE or LEFT/RIGHT each half has its own eight bands; ON and S per half. LINK copies a move on one half to the other.
HUD, ALL BANDS	Edit the selected band, or every band in a table.

Exciter



Up to four bands of saturation, each with its own flavour, on the crossover display. POST FILTER shows where harmonics are being added and puts a high shelf on the added harmonics to tame them. The LED by each band shows how much it is adding.

Control	What it does
MODE	ANALOG: driven odd harmonics. RETRO: a few, decaying slowly. TAPE: brighter odd harmonics. TUBE: harder on attacks. WARM: even harmonics only. TRIODE: subtle tube. DUAL TRIODE: more drive, warmer.
AMOUNT, MIX	How hard the band is driven; how much of the result is blended back.
Oversampling	OFF, 2X or 4X, to keep aliasing down. Each step costs a fraction of a millisecond of latency.

Imager



Stereo width per band on the crossover display, and a vectorscope of the output. Bass in mono below 100 to 150 Hz is the usual mastering move: it makes the low end solid on every system.

Control	What it does
WIDTH	Side level per band: +100 doubles the sides, -100 makes the band mono. AMOUNT scales them all.
STEREOIZE	For mono or narrow sources: a decorrelated copy of the middle goes into the sides so there is something to widen. Mono compatible. MODE I is a delayed copy; MODE II keeps transients sharper. Offered only while a band is being widened.
RECOVER SIDES	When a band is narrowed, what was taken from the sides comes back in the middle instead of being lost. S plays only that. Offered only while a band is being narrowed.
Views	CROSSOVER; CORRELATION over time (red: out of phase); WIDTH per band. Vectorscope: POLAR SAMPLE, POLAR LEVEL or LISSAJOUS, with correlation (+1 the same on both sides, -1 out of phase) and balance.

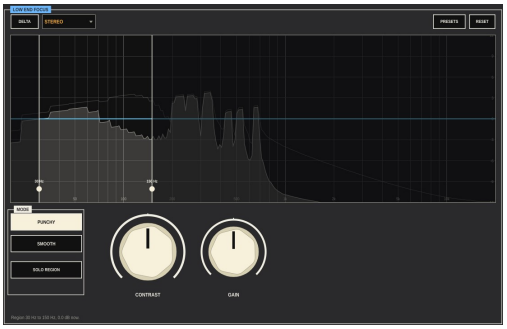
Impact



Microdynamics per band: how far the level moves between a hit and what follows it. One control per band expands or compresses; the traces under the display show the gain in each band over time, up for boost and down for cut.

Control	What it does
IMPACT	Positive opens the microdynamics up (punchier, more open); negative packs them (denser, glued). AMOUNT scales every band.
ENVELOPE	How long an event lasts before the gain returns: milliseconds, or beat divisions of the host tempo with SYNC.
AUTO GAIN, STEREO LINK	Keep the average level where it was; judge both sides from the middle and move them together.

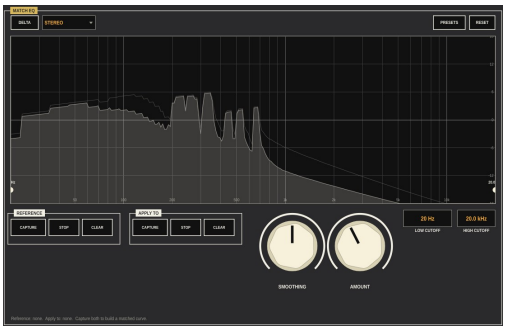
Low End Focus



Contrast between loud and quiet moments in a low region, 20 to 300 Hz (drag the handles). Positive sharpens kick and bass transients and clears the mud under them; negative blurs the low end, the way saturation or a slow compressor would.

Control	What it does
PUNCHY, SMOOTH	Fast, for transients; or slow, for what sustains.
CONTRAST	Positive or negative, as above.
GAIN, SOLO REGION	Make-up on the region; hear the region before processing.

Match EQ



Captures the average spectrum of a reference and of your track and applies the difference as a smoothed, linear-phase curve. Play the reference and CAPTURE under REFERENCE, then STOP; play your track and do the same under APPLY TO. Gold is the reference, blue your track, white the match. The reference snapshot is saved with presets. 43 ms of latency.

Control	What it does
SMOOTHING	Higher smooths the match into a broad shape; lower follows every peak and dip, which is rarely what you want.
AMOUNT	How much of the match to apply. Under 50% is usually wise: the aim is the reference's overall shape, not a copy.
Cutoffs	Below the low cutoff and above the high one the match fades out, so you can match only the low end, say.

Maximizer

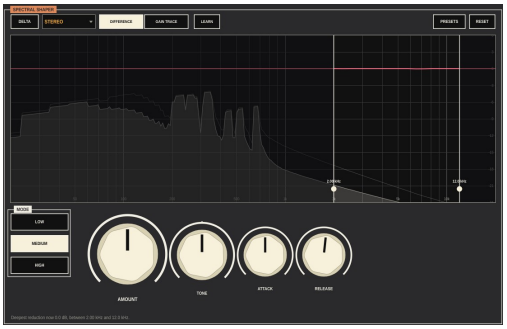


The loudness stage: GAIN into a look-ahead, true-peak limiter with an OUTPUT LEVEL ceiling, plus tools that add loudness so the limiter has less to do. Goes last.

Control	What it does
GAIN, OUTPUT LEVEL	Push the signal into the limiter; the ceiling the output never passes. LINK moves them in opposite directions, so you hear what the limiter does without it getting louder. -1 dB with TRUE PEAK for streaming; -0.1 for a CD.
MODE	LOW LATENCY and I to III are single-band limiters, from the fastest to the most careful. III has STYLES: CLIPPING (loudest, roughest), CRISP, BALANCED, PUMPING. IV pre-limits three bands (CLASSIC, MODERN, TRANSIENT styles); V pre-limits four, each with its own release: the cleanest at high gain.
CHARACTER	Release speed, CLIPPING (0) to VERY SLOW (10). Fast is louder with more distortion; slow is smoother with more pumping. 4 to 6 suits most music.
UPWARD	Upward compression: how far quiet passages may be raised. Peaks stay put, so it adds density without more limiting.
SOFT CLIP	Rounds peaks before the limiter, 4x oversampled. LIGHT starts at -3 dBFS, MODERATE -9, HEAVY -30; AMOUNT is wet/dry.
TRANSIENT EMPH	Sharpens transients before limiting, so drums survive it.

Control	What it does
STEREO INDEPENDENCE	How independently the sides are limited, for transients and sustain separately. 0 keeps the image; 100 is louder but can narrow it.
LEARN GAIN, TARGET	Listens for five seconds and sets GAIN so the output lands near TARGET. A starting point, not a compliance check.
Views	SPECTRUM; GAIN TRACE (limiting from the top, upward boost from the bottom); GR METER per side.

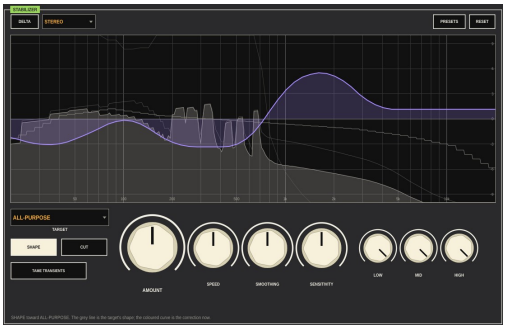
Spectral Shaper



Pulls down narrow peaks that stick out of the spectral envelope, inside an action region (drag the handles; LEARN puts the region around the most resonant part of what is playing). For harshness, whistles and ringing that a static EQ would have to chase. 43 ms of latency.

Control	What it does
MODE	How far above the envelope a peak must be before it is touched: LOW only the worst, MEDIUM, HIGH almost anything.
AMOUNT, TONE	How much of the excess is removed; positive TONE leaves the top brighter, negative darker.
ATTACK, RELEASE	Time to apply a reduction and to let it go. GAIN TRACE shows the deepest reduction over time.

Stabilizer



An adaptive equaliser that corrects toward a tonal-balance target: the assistant's genre profiles, or ASSISTANT, the shape of a reference you gave the assistant. The grey line is the target's shape; the coloured curve is the correction now. 43 ms of latency.

Control	What it does
TARGET, MODE	The balance to aim for. SHAPE boosts and cuts toward it, loudness neutral; CUT only cuts what sticks out above it, which tames resonances without changing the overall balance.
AMOUNT, SPEED	Size of the corrections (at 100 the most is 9 dB); how quickly they follow. Fast is precise but can be heard working.
SMOOTHING, SENSITIVITY	In SHAPE mode, 100 smoothing looks like three or four broad filters, 0 like many narrow ones. Sensitivity: how small a deviation gets corrected.

Control	What it does
TAME TRANSIENTS	React at once to transients.
LOW, MID, HIGH	Scale the correction below 100 Hz, from 100 Hz to 5.6 kHz, and above.

Vintage Compressor



A feedback compressor (its detector listens to its own output) with a filter in the loop, in the manner of the classic bus compressors. Its release is for transients; under sustained compression it lets go much more slowly, so it breathes rather than pumps.

Control	What it does
MODE	SHARP keeps transients crisp; BALANCED judges by the signal; SMOOTH thickens and fills.
THRESHOLD, RATIO	Where it starts, and how much it takes off. The knee depends on the mode.
ATTACK, RELEASE, GAIN	Speed in and out; make-up. AUTO GAIN matches the output level to the input over time.
DETECTION	NONE; a HIGHPASS so bass doesn't pump it; a HIGH SHELF boost or a WIDE BOOST so the highs drive it. S plays what the detector hears.

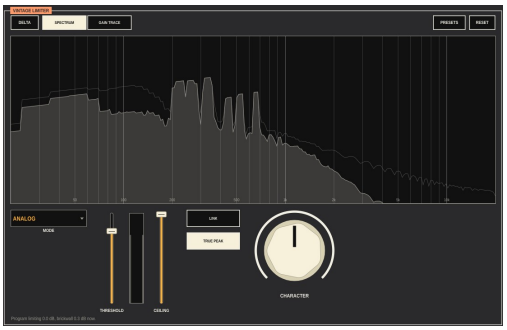
Vintage EQ



Two rows of passive, program-style equalisation with stepped frequencies. The low shelf can boost and cut at once: the boost sits below the cut, so together they give a bump then a dip, the old trick for weight without mud. The high boost has its own bandwidth, which tightens as you boost; the mid peaks widen as they go up.

Control	What it does
Top row	LOW FREQ (20 to 100 Hz), LOW BOOST, LOW CUT; HIGH BOOST FREQ (3 to 16 kHz), HIGH BOOST, BANDWIDTH; HIGH CUT FREQ (5, 10, 20 kHz), HIGH CUT.
Bottom row	LOW-MID (200 Hz to 1 kHz) boost; MID (200 Hz to 7 kHz) cut; HIGH-MID (1.5 to 5 kHz) boost.

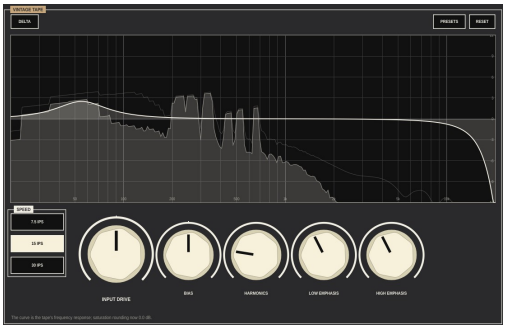
Vintage Limiter



A program limiter in the manner of the variable-mu classics, then a brickwall at the CEILING. Lowering THRESHOLD limits more and makes the master louder: make-up brings the threshold up to the ceiling. An alternative to the Maximizer with more character.

Control	What it does
MODE	ANALOG: fast, thick, tight bass. TUBE: softer, varies with the signal. MODERN: the vintage character into a modern brickwall.
THRESHOLD, CEILING, LINK	As above. LINK moves them together. -0.3 dB ceiling when dithering to 16 bits; -1 for lossy codecs.
TRUE PEAK, CHARACTER	Limit peaks between samples too; attack and release from FAST (0) to SLOW (10).

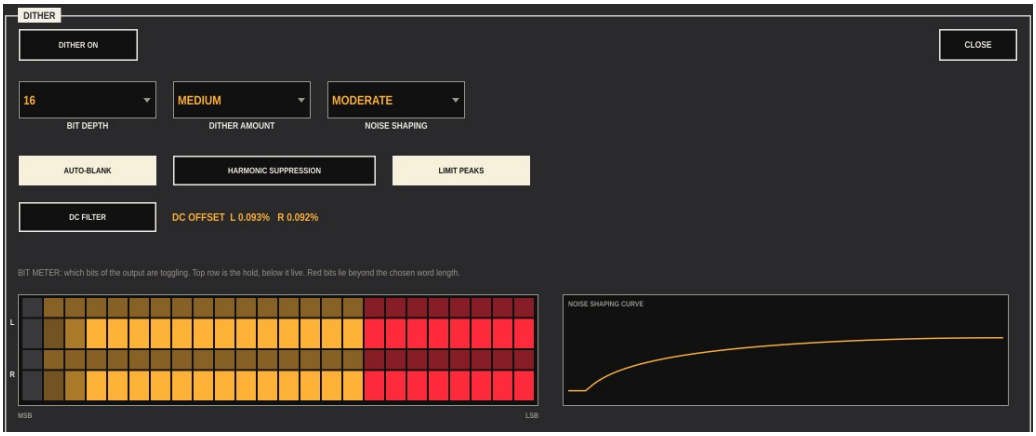
Vintage Tape



The response and saturation of a well-kept two-track tape machine, without the hiss and wobble. The curve is the linear part: the head bump in the lows and how far the top reaches.

Control	What it does
SPEED	7.5, 15 or 30 ips: slower has a bigger, higher head bump and less top end.
INPUT DRIVE	How hard the tape is hit; the level is compensated so only the colour changes.
BIAS	Negative: the highs distort more. Positive: saturation starts earlier, squeezing the dynamics.
HARMONICS	Even-order colour from the electronics.
LOW, HIGH EMPHASIS	The head bump, from flattened to +10 dB; the top end, from rolled off through flat (4) to shimmer.

Dither and delivery



A 24-bit WAV needs no dither. A 16-bit file (CD, some distributors) does: dither replaces the distortion of simply dropping the lowest bits with a tiny, steady noise, and noise shaping moves that noise to where hearing is least sensitive. Dither is the very last thing Burnish does, after the output gain, so Burnish must be the last plugin on the master, and Live's own dither must be off.

Control	What it does
BIT DEPTH	The word length of the file you will export: 16 for CD, 24 for almost everything else.
DITHER AMOUNT	OFF, LOW, MEDIUM (the usual choice) or STRONG, which removes all quantisation distortion at a slightly higher noise floor.
NOISE SHAPING	OFF to MAX: about 14 dB less audible noise at MAX. MODERATE suits most masters.
AUTO-BLANK	Mutes the dither after 0.7 s of digital silence, so fades end in true silence.
HARMONIC SUPPRESSION	With dither OFF only: changes the truncation so its distortion falls away from the music's overtones.
LIMIT PEAKS	Keeps spikes of shaped noise from going over full scale.
DC FILTER	A 1 Hz high-pass just before the first Maximizer or Vintage Limiter in the chain, to remove DC offset. The readout shows how much there is.
BIT METER	Which bits of the output are changing; the top row holds. With dither on, only bits within the chosen depth should light; red ones are beyond it.

Where the song is going

Destination	Loudness	Ceiling	Notes
Streaming	about -14 LUFS	-1.0 dBTP, TRUE PEAK on	Spotify, Apple Music, YouTube and Tidal turn louder songs down anyway; going louder gains nothing and costs punch.
Download, Bandcamp	-11 to -9 LUFS	-1.0 dBTP	Played at the level you give it, so it can sit louder. Still MP3-safe at -1.
CD	-10 to -8 LUFS	-0.3 dBTP	Nothing re-encodes it, so the ceiling can be nearer 0. 16-bit, with dither.

Destination	Loudness	Ceiling	Notes
Vinyl pre-master	-14 to -12 LUFS	-1.0 dBTP	Bass in mono below 150 Hz (Imager), no clipping; the cutting engineer sets the final level.

Exporting from Live

- Select the whole song in the arrangement, from just before the first sound to after the last tail.
- File → Export Audio/Video. Rendered Track: Master. File type WAV (or AIFF); sample rate the session's (44.1 kHz for a CD).
- Bit depth 24 for streaming and downloads, with Burnish's dither off. For a 16-bit CD master: 16 in Live, Burnish's DITHER on at 16, and Live's Dither Options set to No Dither.
- Normalize off: it undoes the ceiling you set.
- Listen to the exported file, start to end, before it goes anywhere.

An album or an EP

Master each song with the same chain and settings as a starting point, so they sound like one record, then adjust per song. Loudness targets measure averages, so a quiet ballad mastered to the same LUFS as a rock song will feel louder than it should: let quieter songs sit 1 to 3 LU lower. Save the chain as a preset (SAVE) so every song starts from it, and use the A B C D snapshots to flick between two songs' settings.

Common mistakes

- **Judging by loudness.** Whatever is louder wins a quick comparison. GAIN MATCH on, always, when you compare.
- **Too many modules.** Each one should be doing something you can hear in its DELTA. If you can't, take it out.
- **Fixing the mix in the master.** A buried vocal or a harsh guitar is easier, and better, fixed in the mix. Go back to the mix if you can.
- **The limiter not last.** Anything after the Maximizer can push a peak over the ceiling. Only dither comes after it.
- **Widening the bass.** Width belongs above 150 Hz or so; below it, mono is solid on every system and cuttable on vinyl.
- **Stacking spectral modules.** Clarity, Stabilizer, Spectral Shaper and Match EQ all react to the same spectrum. Two of them at most, and gently.
- **Normalising on export, or dithering twice.** Normalize off; Live's dither off when Burnish's is on.
- **Mastering on one pair of headphones.** Check on at least two other systems before you call it done.

Latency

Module	Latency at 48 kHz
Clarity, Stabilizer, Spectral Shaper, Match EQ, Equalizer in DIGITAL	43 ms each
Maximizer	about 1 to 5 ms, by MODE (LOW LATENCY least, V most)
Bass Control, Vintage Limiter	about 2 ms
Exciter (2X, 4X), Vintage Tape	under 1 ms
Everything else	none

Burnish tells Live its latency, and Live lines everything up on playback; the I/O panel shows the total. Latency only matters if you play live through Burnish: then use analog EQ, Dynamics and the Maximizer in LOW LATENCY mode.

Where things are kept

Your presets, module presets and assistant targets are kept in ~/Library/Application Support/Burnish (the Presets, Module Presets and Targets folders), with the settings file beside them. OPTIONS → Show settings folder opens it. Copy the folder to move everything to another computer.

Keyboard and mouse

- cmd-Z: undo. Shift-click A B C D: store a snapshot; click: recall it.
- EQ node: drag for frequency and gain, shift-drag to lock an axis, drag the side handles for Q, alt-click to solo its region, double-click to reset, delete to remove; arrow keys nudge (shift coarse, cmd fine).
- Crossover or action-region handle: drag; double-click a crossover to type a frequency.
- Any knob, fader or readout: double-click to reset; scroll to adjust.